

TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

Self-learning smart sensor

Cognitum Seed, a smart sensor that plugs in and learns on its own, is an edge AI device designed to run agent-based AI workloads locally without cloud infrastructure, enabling AI agents to process data, maintain memory, and interact with connected inputs on-device. The system combines local memory, sensor integration, and AI orchestration in hardware form, while supporting stateful agent behaviour through memory functions. The platform includes cryptographic logging to record system activity and support traceability and auditability of AI interactions. Designed for distributed AI operations, Cognitum Seed can coordinate workloads across edge, on-premises, and cloud environments based on deployment needs.

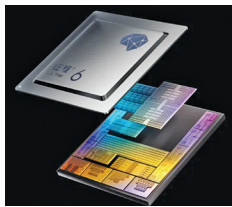
Cognitum One
<https://cognitum.one>



Integrated cockpit and driving vehicle chip

Horizon Robotics has launched Starry, a vehicle chip that integrates cockpit and driving functions on a single chip

for Agentic OS platforms. It supports both driving systems and cockpit computing without separate processors,



replacing the one-function-one-chip structure used in many vehicles and enabling integrated software deployment. The Starry 6 platform runs cockpit AI agents and driver-assistance models on a single computing system,

reduces the need for dual-chip architectures, and lowers per-vehicle hardware costs by about 1500-4000 RMB. It supports millisecond-level latency and high-bandwidth data transfer between workloads.

Horizon Robotics
<https://www.horizon.auto>

Intelligent battery management chip

Texas Instruments (TI) has introduced the BQ79826Z-Q1, an electrochemical impedance spectroscopy (EIS)-enabled



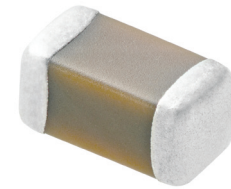
battery monitoring chip for electric vehicles (EVs) and energy storage systems (ESS). The company describes it as a high-cell-count battery monitor, built to deliver predictive intelligence, real-time diagnostics, and visibility into battery cell health. The chip supports monitoring of up to 26 cells per device, enabling up to 44% more channels than previous generations. It also features higher-resolution ADCs, low-noise performance, and EIS measurement speeds up to five times faster than previous solutions. These improvements support more precise battery forecasting and longer usable battery life.

Texas Instruments (TI)
<https://www.ti.com>

Multilayer ceramic capacitor for automotive powertrain and safety systems

Murata Manufacturing has introduced the GCJ21BD72A225KE02, a soft-termination chip multilayer ceramic capacitor (MLCC) for automotive powertrain and safety systems. It is the first soft-termination MLCC to offer

2.2µF at 100V DC in an 0805-inch package. As electric vehicles develop and ADAS and autonomous driving systems expand, PCB space is limited. At the same time, 48V vehicle architectures



require components with high capacitance, high voltage rating, and small size. Mechanical stress

from vibration, thermal cycling, and board flex increases the risk of cracking, underscoring the importance of reliability. This MLCC is designed to address these requirements. Compared to Murata's earlier solution, it reduces mounting area by about 51% and increases capacitance by about 2.2 times compared to its earlier product.

Murata Manufacturing
<https://www.murata.com>

Superchip for personal on-device AI agents

Nvidia has introduced Nvidia RTX Spark, a superchip for personal AI agents on Windows PCs. It shifts computing from tool-based use to systems that act like AI



assistants. The platform is designed for up to 1 petaflop of AI compute and 128GB unified memory for local AI tasks. Nvidia and Microsoft are building a Windows framework for on-device AI agents with security features for identity, isolation, and policy control. It includes Nvidia OpenShell runtime for agent control, privacy routing, and data protection. This allows agents to run locally, use applications, manage tasks, create content, and search files with user control.

Nvidia Corporation
<https://www.nvidia.com/en-in>